TSITA MOGAU SETSHABA PHIAS

Computer Science Graduate

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Portfolio Linkedin Github

SUMMARY

Computer Science and Statistics graduate currently completing an Honours degree in Computer Science at the University of Limpopo. Skilled in software development, computer networking, cybersecurity, database management, data analytics, and artificial intelligence. Experienced in Python, Java, JavaScript, SQL, Ubuntu Linux, and machine learning through academic and research projects. Developed practical solutions involving Linux router configuration, database systems, DDoS attack detection, and predictive modelling. Strong analytical, problem-solving, and technical troubleshooting skills, with a passion for developing reliable technology solutions and continuously learning new technologies.

EDUCATION

02/2026 – Present
Polokwane, Mankweng

University of limpopo
Bachelor of Science Honours in Computer Science, Relevant Modules: Advanced Database Systems, Computer Networking, Operating Systems, Computer Security.
Current Academic Average: 70%

02/2022 – 12/2025
Polokwane, Mankweng

University of Limpopo
Bachelor of Science in Computer Science & Statistics, Relevant Modules: Data Structures, Computer Architecture, Operating Systems, Database Systems, Artificial Intelligence, Computer Networking.
Final Academic Average: 60%

PROJECTS

02/2025 – 05/2025

Database Management
- Designed a centralized database system to manage projects, employees, clients, inventory, and IT support tickets, including ER diagrams, a normalized schema, and data dictionary.
- Developed SQL queries to support key reporting and business operations.

02/2026 – 11/2026

Honours Research Project
Design and Implementation of an Interpretable Transformer-Based Framework for Long-Term Cardiovascular Mortality Risk Prediction.
- Developed CardioTransNet-I, a transformer-based machine learning model to predict 10-year cardiovascular mortality using data from 142,244 participants.
- Applied data preprocessing, statistical modelling, cross-validation, and SHAP explainable AI to evaluate predictions and identify key risk factors.

02/2026 – 05/2026 **Ubuntu Linux Router Configuration**

- Configured an Ubuntu Linux router with NAT, DHCP, firewall, and routing capabilities.
- Created two virtual machines to test network connectivity and communication through the router.

02/2025 – 11/2025 **End-to-end model for xitsonga language**

Machine Learning Project

- Self-taught Hugging Face Transformers to fine-tune a speech recognition model for a low-resource language, achieving 10% WER on the NCHLT dataset.
- Gained end-to-end ML pipeline experience, from data cleaning to deployment.

08/2026 – 10/2026 **Computer Security Research Project**

Detection and Ranking of Suspected Botmaster Nodes During DDoS Attacks

- Collaborated on a cybersecurity research project using Random Forest to detect DDoS attacks, achieving 99.93% accuracy.
- Analysed network traffic using CICDDoS2019 and CTU-13 datasets to rank suspected botmaster nodes.

SKILLS

Database

•PostgreSQL
•Oracle SQL
•Firebase
•MongoDB

IT Support & Documentation

- Git
- Github desktop
- Hugging Face
- analytical skills
- Power BI

Technical

- SQL (advanced joins, window functions, query optimization)
- ETL/ELT pipeline development
- Data warehousing (star schema, dimensional modeling)
- Python (pandas)
- PostgreSQL/MySQL
- MongoDB
- Git

Networking

- TCP/IP fundamentals
- Network protocols (DNS, DHCP, HTTP/HTTPS, FTP)
- OSI Model understanding
- Router and switch configuration basics
- Network troubleshooting
- IP addressing and subnetting
- PowerShell
- LAN / WAN / Wi-Fi
- Ubuntu Linux

Data Analysis tools

- SPSS Program
- SAS Program
- Microsoft Excel
- B-BBEE

Artificial Intelligence

- Machine Learning
- Transformers
- Predictive Modelling

Programming Languages

- HTML
- CSS
- REACT
- JAVASCRIPT
- Python
- JAVA

Software Development

- Data Structure & Algorithms
- Object-Oriented Programming
- Git/GitHub
- REST APIs
- Oracle APEX

Data Science

- Data Cleaning
- Data Preprocessing
- Statistical Analysis

ACHIEVEMENTS/CERTIFICATES

Network Micro-Certification.
Completed Network Layer Protocols Micro-Certification with comprehensive understanding.

Hackathon competition Certificate.
Led team through 3-day Hackathon competition, applying technical problem-solving and collaboration skills in a fast-paced environment.

Digital Literacy Made Easy - NEMISA.
Certificate of Attendance for Introduction to Digital Literacy, Microsoft Office and Emerging Technologies.

ULSTEAM Ambassador Programme – AI & Robotics | September 2026

- Participated in a 5-day AI and machine learning workshop, Explored data analytics and Edge AI for flood early warning.

REFERENCES

Mr Maluleke, *Data Analysis Lecture,*
University of Limpopo
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Dr MS Madiba, *Honours Research Supervisor,*
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Dr Baloyi, *Hackathon Organiser,*
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LANGUAGES

Sepedi

English